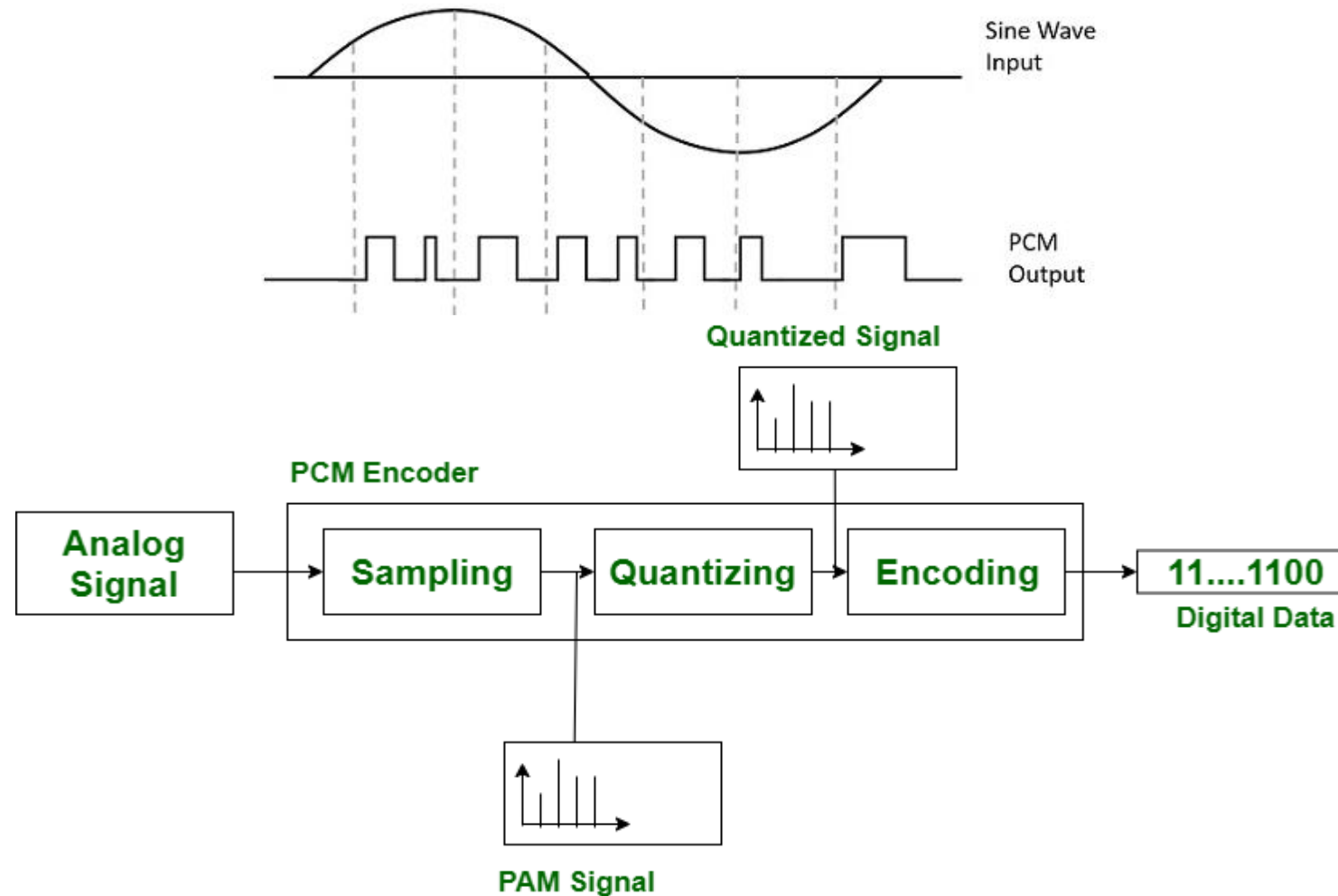


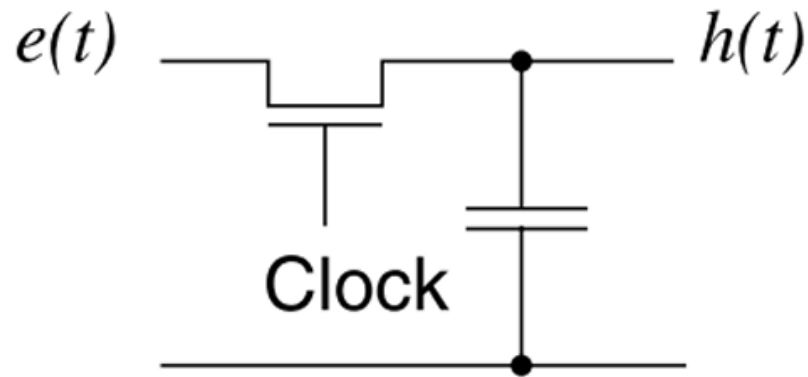
PULSE CODE MODULATION

Pulse Code Modulation or PCM converts the analog information into a binary sequence (1 and 0).



SAMPLING

It is the process by which, we convert continuous time signal into discrete time signal by taking the signal values at some distinct points in time. So, a sample is the numeric value of an analog signal at a specific time making it a discrete time signal.



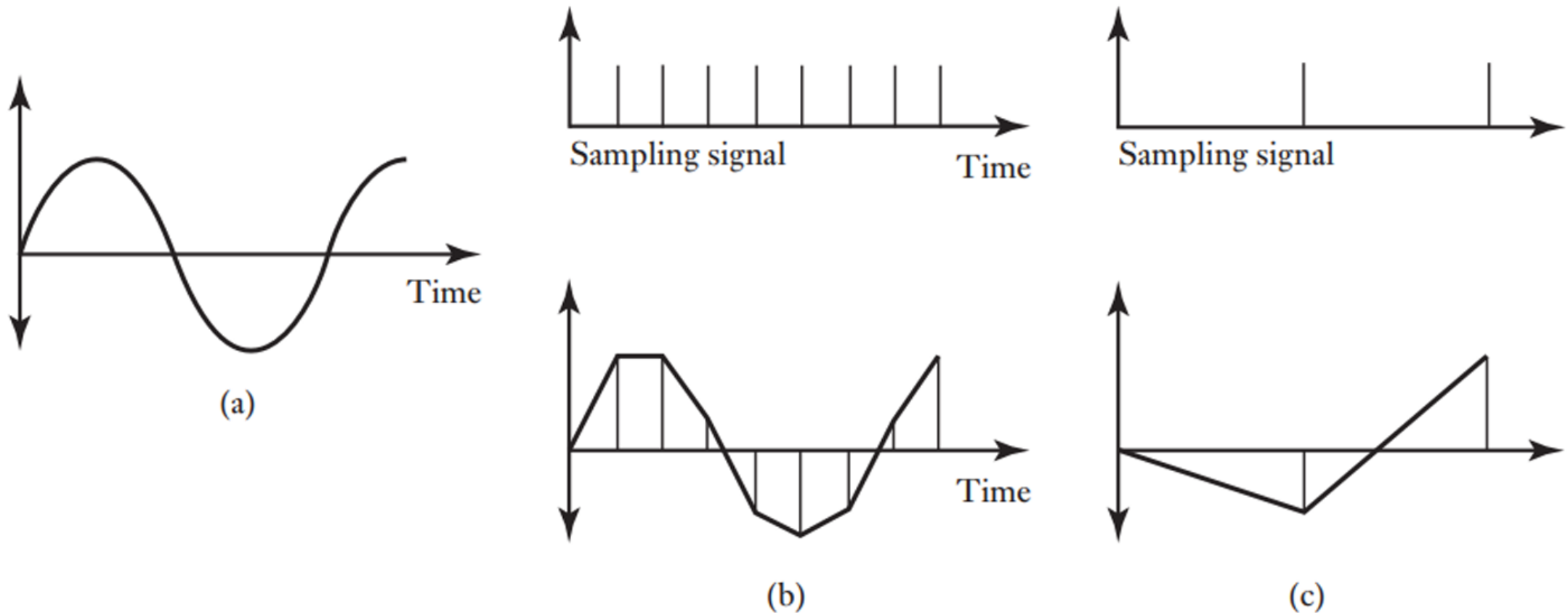
Sample and Hold Circuit

$$\text{Sampling rate} = \frac{1}{T} = f \text{ (Hz)}$$

SAMPLING THEOREM

4

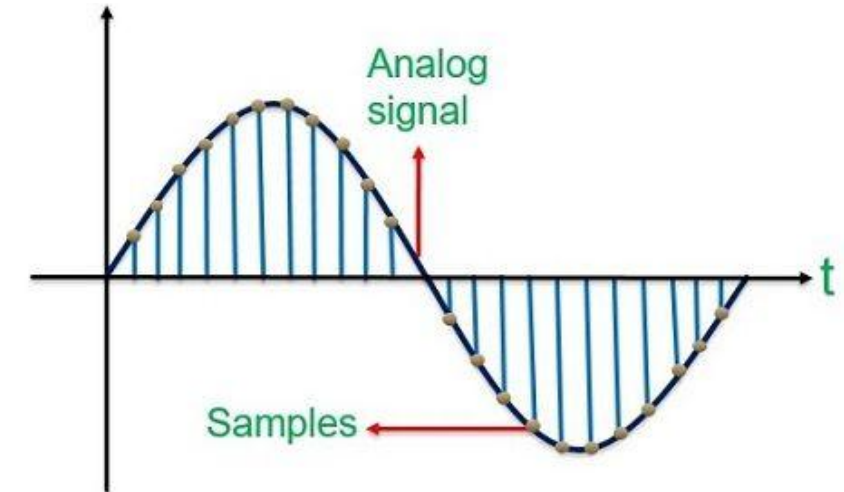
According to Nyquist criterion, the sampling rate must be twice as the highest frequency of the analog signal. Otherwise, reconstructed signal will represent false image known as **Aliasing**.



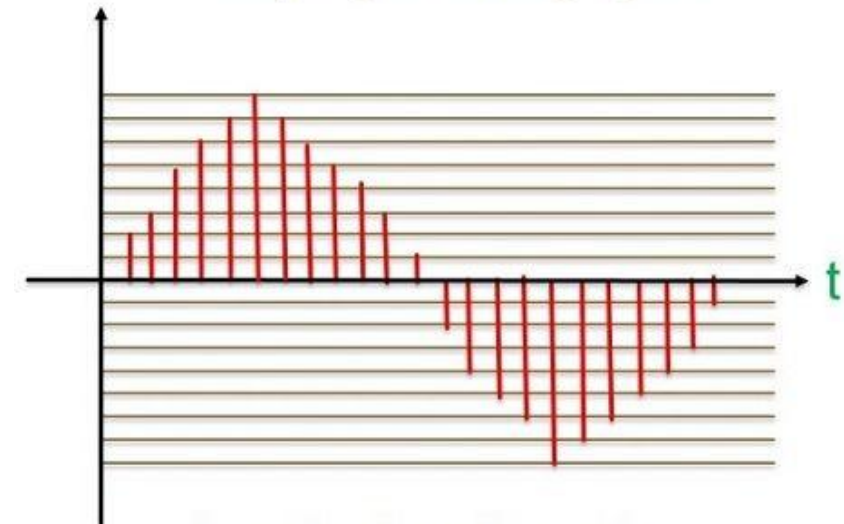
QUANTIZATION

It is the process to represent a continuous-valued signal with a limited set of discrete values. This done because although the time is discrete after sampling, the amplitude is still continuous value.

- Amplitude range of the signal is divided into L discrete levels
- Each sample is rounded off to the nearest level
- More level= less noise and distortion
- 2^N quantization level for N – bit quantizer



Sampling of analog signal



Quantization of samples

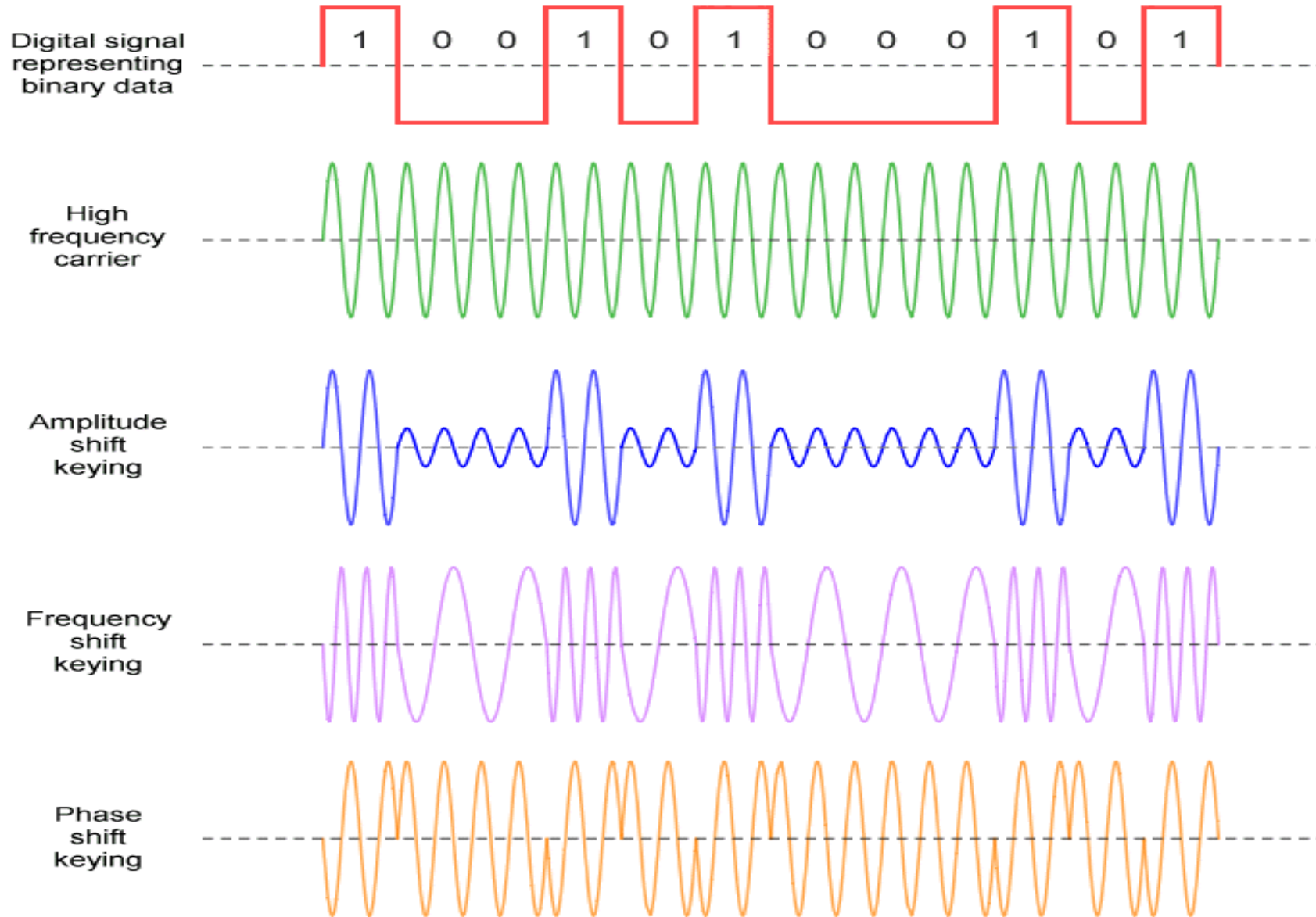
ENCODING

The process of converting each quantized level into binary code is called **Encoding**.

If no. of quantization level= L
Then, no. of bits per sample, $n = \log_2 L$

For example,
 $L=8$ and $n=3$ bits
Then, levels encoded 000, 001, 010.....111

ASK, FSK & PSK





THANK YOU

